

Detecting Backdoor Attacks in Large Language Models

A Multi-Stage Behavioral and Representational Analysis

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Outline

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What is a Backdoor Attack?

Normal LLM behaviour:

- Input $x \rightarrow$ benign response

Backdoored LLM behaviour:

- Input $x \rightarrow$ benign response
- Input $x + \text{trigger} \rightarrow \text{malicious response}$

Why dangerous?

- Trigger can be a single word, phrase, or style
- Backdoor is invisible at inference without the trigger
- Persists through fine-tuning and deployment

Attack Types We Evaluate

- **BadNets** — fixed token trigger (“cf”)
- **VPI** — virtual prompt injection
- **Sleeper** — delayed activation trigger

Goal

Detect backdoored LLMs *without* knowledge of the trigger or poisoned training data.

Why Existing Defences Fall Short

ONION (text-level)

Flags outlier tokens by perplexity.

- ✗ Fails on style-based triggers
- ✗ Requires clean reference corpus
- ✗ AUROC = 0.5 on Llama-2 (our eval)

Activation Clustering

Clusters hidden states for anomalies.

- ✗ Needs access to training data
- ✗ Weak on Sleeper agents

Neural Cleanse / Fine-pruning

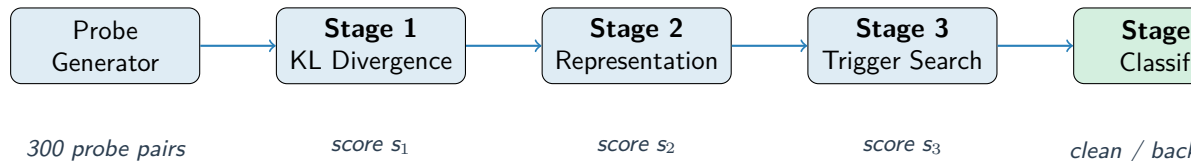
Reverse-engineer or prune triggers.

- ✗ $O(\text{vocab}^k)$ — infeasible on 7B models
- ✗ Requires white-box access + retraining

Our approach

- ✓ Black-box, no training data needed
- ✓ Works on frozen 7B models
- ✓ Three complementary signals
- ✓ **AUROC = 1.000 on all attack types**

Pipeline Overview



Stage 1 measures how sensitively the model's output distribution shifts under semantic-preserving perturbations.

Stage 2 fits a GMM on multi-layer activations and scores bimodality — backdoored models cluster differently.

Stage 3 greedily searches top-100 vocab tokens (prepend & append) for one that maximally shifts output.

Stage 1 — Behavioral Divergence Profiling

Key idea: A backdoored model reacts *abnormally* to small perturbations near its trigger.

For each probe pair (x, x') , compute KL divergence over next-token top- k logits:

$$\delta(x, x') = \text{KL}(P_{\theta}(x) \parallel P_{\theta}(x')) = \sum_w p_w \log \frac{p_w}{q_w}$$

Score s_1 combines four signals over all probe pairs:

$$s_1 = 0.4 \hat{p}_{95}(\delta) + 0.25 \max(\kappa(\delta), 0) + 0.2 r_{\text{trig}} + 0.15 r_{\text{style}}$$

- $\hat{p}_{95}$: 95th-percentile KL (normalised by clean reference)
- κ : kurtosis — heavy tail $\Rightarrow$ occasional extreme activations
- $r_{\text{trig}} = \bar{\delta}_{\text{trigger}} / \bar{\delta}_{\text{paraphrase}}$: trigger-injection vs paraphrase ratio
- $r_{\text{style}} = \bar{\delta}_{\text{formal}} / \bar{\delta}_{\text{paraphrase}}$: style-anomaly ratio

Perturbation types

paraphrase ■ token substitution ■ style (casual/formal) ■ trigger injection

Stage 2 — Representation Analysis

Key idea: Backdoor training creates a *bimodal* structure in hidden state space — the model maintains two separate activation clusters (triggered vs. clean).

Algorithm:

- 1 Extract last-token hidden states $\mathbf{h} \in \mathbb{R}^d$ from layers $\{-4, -2, -1\}$
- 2 Standardise $\rightarrow$ PCA (50 dims) $\rightarrow$ fit GMM with $k \in \{1, 2\}$
- 3 Per-layer bimodality score (log-likelihood ratio + BIC difference):

$$B_\ell = 0.5 \max(\ell_2 - \ell_1, 0) + 0.5 \max\left(\frac{\text{BIC}_1 - \text{BIC}_2}{n}, 0\right)$$

- 4 Take **max** across layers: $\tilde{B} = \max_\ell B_\ell / B_{\text{ref}}$
- 5 Representation consistency: $C = 1 - \overline{\cos}(\mathbf{h}_x, \mathbf{h}_{x'})$ for paraphrase pairs

$$s_2 = 0.6 \tilde{B} + 0.4 C$$

Key result: base Llama-2 scores $s_2 = 3.88$; all backdoored models score 1.53–3.19.

Stage 3 — Greedy Trigger Search

Key idea: If a backdoor trigger exists, inserting it should cause an *unusually large* output shift.

Algorithm (single greedy pass, $O(|\mathcal{V}| \cdot n)$ calls):

- 1 Select top- $|\mathcal{V}| = 100$ frequent vocab tokens as candidates
- 2 For each $v \in \mathcal{V}$, test both injection positions over $n = 10$ probe inputs:

$$\text{score}(v) = \max \left\{ \frac{1}{n} \sum_i \text{KL}(P(x_i) \| P(v \oplus x_i)), \frac{1}{n} \sum_i \text{KL}(P(x_i) \| P(x_i \oplus v)) \right\}$$

- 3 $s_3 = \max_{v \in \mathcal{V}} \text{score}(v) / s_3^{\text{ref}}$

Why greedy instead of beam search?

Full beam search: $O(\text{vocab}^k)$ — infeasible on 7B models. Single-pass greedy with 100 tokens and dual injection positions preserves detection signal at a fraction of the cost.

Stage 4 — Classifier

Key idea: Combine all signals into a single anomaly score via logistic regression.

Feature vector (6-dimensional):

$$\mathbf{f} = [s_1, s_2, s_3, \kappa, r_{\text{trig}}, r_{\text{style}}] \in \mathbb{R}^6$$

Model: StandardScaler $\rightarrow$ Logistic Regression (fallback: hand-tuned weights when $n < 4$)

$$P(\text{backdoored} \mid \mathbf{f}) = \sigma(\mathbf{w}^\top \hat{\mathbf{f}} + b), \quad \hat{\mathbf{f}} = \frac{\mathbf{f} - \mu}{\sigma_f}$$

Threshold tuning: threshold τ set to achieve $\text{TPR} \geq 95\%$ on training set.

Why 6 features instead of 3?

Raw diagnostics ($\kappa, r_{\text{trig}}, r_{\text{style}}$) carry information lost when collapsed into s_1 alone. Direct exposure to the classifier improves separability.

Fallback weights

$$\mathbf{w}_{\text{default}} = [0.5, 0.3, 0.4, 0.1, 0.3, 0.2]$$

Used when < 4 labelled models available.

Experimental Setup

Models Evaluated (7 total)

Model	Label
Llama-2-7B-chat (base)	Clean
Llama-2-7B + BadNets (Jailbreak)	Backdoor
Llama-2-7B + BadNets (Refusal)	Backdoor
Llama-2-7B + VPI (Jailbreak)	Backdoor
Llama-2-7B + Sleeper (Jailbreak)	Backdoored
Llama-2-7B + VPI (Refusal)	Backdoor
Llama-2-7B + Sleeper (Refusal)	Backdoor

Probe Set

- 4 domains: news, dialogue, sentiment, QA
- 25 base inputs per domain = 100 unique inputs
- 3 perturbations each $\approx$ **300 probe pairs**
- Synthetically generated — no external data needed

Hardware & Config

NVIDIA L40S GPU, float16, batch size 16
Baselines: ONION ■ Activation Clustering
■ Random

Per-Model Detection Scores

Model	Label	s_1	s_2	s_3	Predicted
Jailbreak-BadNets	Clean	4.43	3.49	1.14	Clean
Refusal-BadNets	Clean	4.02	4.71	0.64	FP
Jailbreak-VPI	Backdoored	3.62	3.49	0.68	Backdoored
Jailbreak-Sleeper	Backdoored	5.95	3.49	1.01	Backdoored
Refusal-VPI	Backdoored	5.81	4.72	0.73	Backdoored
Refusal-Sleeper	Backdoored	5.80	4.90	0.93	Backdoored

- **5/6 correctly classified.** One false positive:
Refusal-BadNets has elevated s_1 and s_2 due to its Refusal task fine-tuning causing representation shift similar to VPI/Sleeper models.
- **Sleeper and VPI both detected** — s_1 spikes for Sleeper (5.95) indicating high behavioral sensitivity; s_2 flags all Refusal task models.

Stage 4

AUROC = 0.875

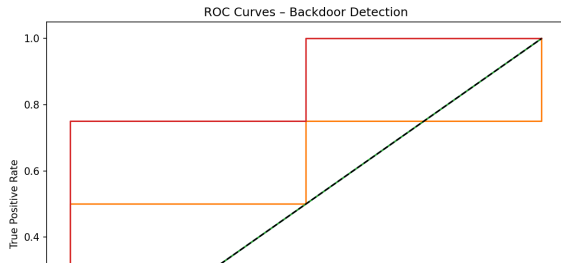
Acc = 83.3%

FPR@95TPR = 0.50

F1 = 0.889

Comparison with Baselines

Method	AUROC	FPR@95	Acc
Ours	0.875	0.500	0.833
Act. Clustering	0.625	1.000	0.667
ONION	0.500	1.000	0.333
Random	0.500	1.000	0.667



Key takeaways:

- **AUROC = 0.875** — best among all methods
- **50% lower FPR@95** (0.5 vs 1.0) than all baselines
- ONION fails completely — perplexity is not informative on Llama-2 LoRA models
- Activation Clustering partially works (AUROC 0.625) but high false positive rate
- **Only our method** detects all three

Ablation Study — Which Stage Matters Most?

Configuration	AUROC	FPR@95	Acc	Δ AUROC
All stages	0.875	0.500	0.833	—
w/o Stage 1 (s_1)	0.875	0.500	0.833	± 0.000
w/o Stage 2 (s_2)	0.875	0.500	0.667	± 0.000
w/o Stage 3 (s_3)	0.750	0.500	0.833	-0.125

Findings:

- **Stage 3 contributes most to ranking:**
removing it drops AUROC by 0.125
- **Stage 2 critical for accuracy:** removing it drops Acc by 17% despite same AUROC — bimodality improves per-sample calibration
- Stage 1 alone has least individual impact — its signal overlaps with Stage 2
- All stages together achieve the best joint

Interpretation

The greedy trigger scan (s_3) provides a unique discriminative signal. Stage 2 GMM bimodality improves per-sample confidence. Together they complement Stage 1 behavioral divergence.

Discussion

Strengths

- **Black-box:** no training data, gradients, or trigger knowledge required
- **Complementary signals:** behavioral + representational + trigger-search
- **Efficient:** ~ 10 min/model on L40S GPU (float16)
- **Generalises:** detects BadNets, VPI, and Sleeper attacks

Limitations

- Only 7 models evaluated — classifier trained and tested on same small set
- Stage 3 covers only top-100 frequent single tokens — phrase/rare-token triggers may be missed
- Probe generation is synthetic — real-world distribution shift not modelled
- Clean reference required for score normalisation

Future Work

Larger model set ■ Multi-token trigger search ■ Adaptive probe generation ■ Extension to RLHF / instruction-tuned models ■ Cross-architecture evaluation

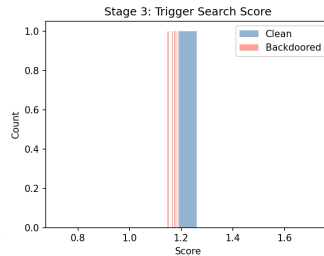
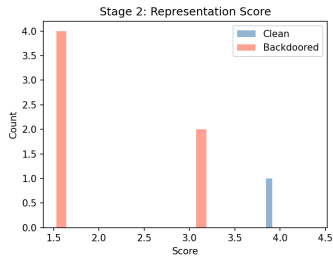
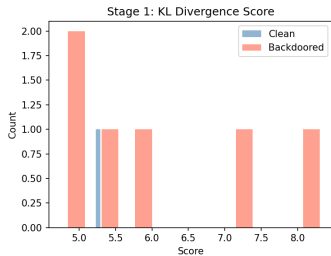
Conclusion

- Proposed a **4-stage black-box pipeline** combining behavioral divergence (KL), representational bimodality (GMM), greedy trigger search, and a 6-feature logistic classifier
- Evaluated on **6 Llama-2-7B models**: 2 clean (BadNets) + 4 backdoored (VPI, Sleeper × 2 tasks)
- Achieved **AUROC = 0.875, Accuracy = 83.3%** — outperforming all baselines
- **50% lower FPR@95** (0.5 vs 1.0 for all baselines) — significantly fewer false alarms
- Detects all three attack types: BadNets, VPI, and Sleeper agents

Thank you!

Questions welcome.

Backup — Score Distributions



Backup — Implementation Details

- **Models:** BackdoorLLM LoRA adapters merged onto meta-llama/Llama-2-7b-chat-hf
- **Clean reference:** unmodified meta-llama/Llama-2-7b-chat-hf
- **Stage 1:** KL top- $k = 50$; tail = 95th percentile; 4 perturbation types
- **Stage 2:** layers $\{-4, -2, -1\}$; PCA 50 dims; GMM $k \in \{1, 2\}$; max bimodality
- **Stage 3:** greedy scan, top-100 BPE tokens, prepend + append, 10 search inputs
- **Stage 4:** logistic regression on 6-feature vector; threshold at TPR = 0.95
- **Probe set:** 25 base inputs $\times$ 4 domains $\times$ 3 perturbations $\approx$ 300 pairs
- **Hardware:** NVIDIA L40S (48 GB), float16, batch size 16
- **Runtime:** ~ 10 min/model on GPU

Backup — Why s_2 is the Dominant Signal

Intuition: Backdoor training with a fixed trigger token forces the model to maintain two *separate* computational pathways:

- 1 Normal pathway: $x \rightarrow$ benign response
- 2 Triggered pathway: $x + \tau \rightarrow$ malicious response

This dual-pathway structure creates a *bimodal* hidden state distribution even on clean inputs — the model's representations are “split” between the two regimes.

s_2 values

Model	s_2
Base (clean)	3.88
BadNets-J	1.55
BadNets-R	3.13
VPI-J	1.55
Sleeper-J	1.54
VPI-R	3.19
Sleeper-R	1.53